

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Organic Hydroxy Compound Transport (GO:0	0.24180952	35	7.497e-07	4.039e-03	SLC10A2:36 SLC51A:275 ATP8B1:293 ABCD3:360 RALBP1:374 ABCG5:389
Monocarboxylic Acid Transport (GO:001571	0.16677667	61	6.786e-06	1.828e-02	SLC5A12:3 SLC16A5:31 SLC10A2:36 SLC22A13:38 SLC6A12:42 FABP1:90
Fatty Acid Catabolic Process (GO:0009062	0.16479050	59	1.447e-05	2.599e-02	EHHADH:5 EC12:192 ABHD3:241 ABCD3:360 ACADM:409 NUDT7:483
Alpha-Amino Acid Biosynthetic Process (G	0.36508316	10	6.404e-05	8.624e-02	SDS:55 BCAT2:145 CPS1:339 SDSL:55 ILVBL:1618 BCAT1:2155
Nitrogen Metabolic Process (GO:0006067)	0.99570009	8	1.064e-04	9.551e-02	SULT1A2:166 SULT1A3:175 SULT1E1:215 SULT1B1:1195 ACS2:52 RHAD:41735
Ethanol Compound Transport (GO:0071705)	0.09433085	144	9.741e-05	9.551e-02	SLC22A13:38 SLC6A6:46 RHAG:62 RHD:163 RHC2:182
Monocarboxylic Acid Metabolic Process (G	0.12152510	80	1.752e-04	1.348e-01	AGXT:45 CYP1A1:47 ABCB11:103 HOGA1:131 NPC1:159 CPT1A:176
Acid Secretion (GO:0046717)	0.35812655	8	4.522e-04	1.503e-01	DRD3:249 SLC51A:275 SLC51B:808 UGT1A6:1638 SLC22A16:2035 SLC26A6:2569
Alpha-Amino Acid Catabolic Process (GO:1	0.19364370	29	3.089e-04	1.503e-01	TAT:30 GOT2:119 HOGA1:131 MCCC1:260 SARDH:271 HIBCH:287
Amino Acid Catabolic Process (GO:0009063	0.16864423	38	3.241e-04	1.503e-01	GOT2:119 HOGA1:131 MCCC1:260 SARDH:271 HIBCH:287 BCKDK:1066
Bile Acid And Bile Salt Transport (GO:00	0.25978112	16	3.219e-04	1.503e-01	SLC10A2:36 SLC51A:275 ATP8B1:293 ABCD3:360 SLCOI1:394 SLC10A6:801
Branched-Chain Amino Acid Metabolic Proc	0.26181456	15	4.477e-04	1.503e-01	BCAT2:145 MCCC1:260 HIBCH:287 BCKDK:1066 PCCB:1501 IVD:1573
Fatty Acid Metabolic Process (GO:0006631	0.09670180	108	5.301e-04	1.503e-01	CYP1A1:47 ABCB11:103 CPT1A:176 PLA2G1B:260 ABCD3:360 ACADS:390
Intracellular Protein Transport (GO:0006	-0.06619845	265	2.238e-04	1.503e-01	RPAIN:22 STX3:60 SCFD2:94 STX4:134 TIMM21:135 SYNRR:238
Lipid Translocation (GO:0034204)	0.22126556	21	4.495e-04	1.503e-01	ATP8B1:293 ABCB1:449 ABCC1:683 ATP9B:1336 ABCB4:1426 TMEM30A:1800
Organic Cation Transport (GO:0015695)	0.18088109	31	4.941e-04	1.503e-01	SLC6A12:42 SLC44A4:285 SLC22A4:341 RALBP1:374 TSPD:497 SLC22A1:509
Organonitrogen Compound Biosynthetic Pro	0.09593978	110	5.221e-04	1.503e-01	SPTLC3:49 ACER1:86 ELOVL7:107 BCAT2:145 SULT1A2:166 ABHD3:241
Phospholipid Translocation (GO:0045332)	0.19390350	27	4.901e-04	1.503e-01	ATP8B1:293 ABCB1:449 ABCC1:683 ANO6:975 ATP9B:1336 ABCB4:1426
Protein Transport (GO:0015031)	-0.06376495	255	4.855e-04	1.503e-01	STX3:60 MCOLN1:69 SCFD2:94 STX4:134 TIMM21:135 TVP2C3:157
L-phenylalanine Catabolic Process (GO:00	0.43205456	5	8.204e-04	1.594e-01	TAT:30 HGD:158 GSTZ1:456 QDPR:2197 IL4I1:2249 NA
L-phenylalanine Metabolic Process (GO:00	0.43205456	5	8.204e-04	1.594e-01	TAT:30 HGD:158 GSTZ1:456 QDPR:2197 IL4I1:2249 NA
Amino Acid Transmembrane Transport (GO:0	0.15220647	41	7.515e-04	1.594e-01	SLC6A6:46 SLC7A9:317 SLC22A4:341 SLC6A14:468 SLC38A4:648 SLC6A20:793
Calcium-Dependent Cell-Cell Adhesion Via	-0.18892765	26	8.583e-04	1.594e-01	CDH7:218 CDH22:543 CDH12:638 CDH2:684 CDH23:1236 CDH18:1244
Embryonic Skeletal/ System Development (G	-0.19319556	25	8.302e-04	1.594e-01	KIAA1217:30 HOGA4:118 WNT11:398 WNT5A:603 HOXD4:489 HOXB3:1015
Erythrose 4-Phosphate/ Phenophenolpyruvat	0.43205456	5	8.204e-04	1.594e-01	TAT:30 HGD:158 GSTZ1:456 QDPR:2197 IL4I1:2249 NA
Fat-Soluble Vitamin Metabolic Process (GO	0.22720074	18	8.486e-04	1.594e-01	CYP11A1:27 CYP1A1:47 CYP2R1:264 NQO1:663 LRAT:884 GGCCX:1132
Fatty Acid Beta-Oxidation (GO:000635)	0.14118681	47	8.195e-04	1.594e-01	EHHADH:5 GCDH:11 EC12:192 ABCD3:360 ACADM:409 ACAD11:498
Glucoseneogenesis (GO:0006094)	-0.18436463	29	5.928e-04	1.594e-01	ENO3:463 PPARGC1A:1038 SLC25A12:1139 DMH2:1289 ENO2:1330 GPI:1363
Quaternary Ammonium Group Transport (GO:	0.12929673	12	6.907e-04	1.594e-01	SLC44A4:285 SLC22A4:341 SLC22A1:509 SLC6A12:42 SLC44A5:1503 SLC5A7:2291
Lipidized Translocation (GO:0006869)	0.10249030	88	9.070e-04	1.611e-01	EHHADH:5 SLC10A2:36 FABP1:90 ABCB11:103 GOT2:119 MTP:157
Polarized Epithelial Cell Differentiation	0.27539879	12	9.567e-04	1.611e-01	RHOA:238 CDC42:328 MSN:1164 SCRBB:1858 OPHN1:2204 MYO9A:2210
Vitamin D Metabolic Process (GO:0042359)	0.33797007	8	9.325e-04	1.611e-01	CYP1A1:47 CYP2R1:264 CYP2A1:1268 UGT1A6:1638 LGMN:2996 CYP27B1:3786
Aromatic Amino Acid Family Catabolic Proc	0.30000083	10	1.020e-03	1.656e-01	TAT:30 HGD:158 GSTZ1:456 QDPR:2197 IL4I1:2249 KMO:3496
Monocarboxylic Acid Catabolic Process (G	0.21139565	20	1.068e-03	1.692e-01	AGXT:45 HOGA1:131 PPARD:1224 LPIN2:1521 CYP26C1:1914 IDUA:2503
Amino-Acid Betaine Transport (GO:0015838	0.35246541	7	1.241e-03	1.853e-01	SLC6A12:42 SLC22A4:341 SLC6A20:793 SLC22A16:2035 SLC22A5:2355 SLC25A29:3821
Cytoplasmic Translation (GO:0002181)	0.14897708	39	1.294e-03	1.853e-01	RPL34:56 RPS27A:72 UBAS2:94 RPL17:178 RACK1:179 RPS15A:208
Regulation Of Monoatomic Ion Transmembra	0.17557719	28	1.307e-03	1.853e-01	KCNJ1:43 KCNJ13:611 KCN9J:639 KCNJ16:881 GRIN2B:1038 GRIN2A:1086
Steroid Metabolic Process (GO:0008202)	0.11967747	61	1.241e-03	1.853e-01	CYP11A1:27 DHRS4:39 CYP1A1:47 CYP17A1:79 NPC1:159 NR12:201
Fatty Acid Derivative Biosynthetic Proce	0.18370453	25	1.481e-03	1.966e-01	GCDH:11 FAR2:135 GGTLC3:641 SLC27A5:840 ACS35:865 GSLC1:922
Neurotransmitter Transport (GO:0006836)	0.13398357	47	1.496e-03	1.966e-01	SNAP25:67 SLC1A1:74 SYT1:447 SLC22A1:509 UNC13A:525 SLC17A7:1109

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
HSIAO_LIVER_SPECIFIC_GENES	0.13392857	196	1.106e-10	7.176e-07	ALB:9 TAT:30 AGXT:45 SDS:55 ALDH1A1:65 FABP1:90
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.07247723	624	8.389e-10	2.348e-06	SLC5A12:3 ALB:9 SLC14A2:15 SLC4A1:20 CAMK2B:21 SLC6A12:42
REACTOME_METABOLISM_OF_LIPIDS	0.07054983	651	1.086e-09	2.348e-06	EHHADH:5 ALB:9 CYP11A1:27 SLC10A2:36 CYP1A1:47 SPTLC3:49
CARRILLOREIXACH_HEPATOBLASTOMA_VS_NORMAL	0.05878053	911	2.635e-09	4.274e-06	EHHADH:5 HMCE5:7 GCDH:11 CAMK2B:21 CYP11A1:27 TAT:30
REACTOME_METABOLISM_OF_AMINO_ACIDS_AND_D	0.09701045	299	8.844e-09	1.147e-05	GCDH:11 TAT:30 GNM1:35 SLC6A12:42 AGXT:45 SDS:55
REACTOME_BIOLOGICAL_OXIDATIONS	0.12836463	146	8.987e-08	7.931e-05	MGST3:18 CYP11A1:27 CYP1A1:47 ALDH1A1:65 CYP2F1:68 CHAC2:83
REACTOME_DNA_METHYLATION	0.29206290	28	8.877e-08	7.931e-05	H4C2:156 H4C1A:207 H3C13:273 H4C15:325 H2BC15:402 H4C3:448
REACTOME_FATTY_ACID_METABOLISM	0.12426240	155	9.781e-08	7.931e-05	EHHADH:5 CYP1A1:47 ELOVL7:107 SCD:1118 CPT1A:176 EC12:192
WOO_LIVER_CANCER_RECURRENCE_DN	0.17651195	66	7.189e-07	5.182e-04	GCDH:11 ACY1:104 GOT2:119 MTP:157 HGD:158 SLC2A9:258
WP_NUCLEAR_RECEPTORS_IN_LIPID_METABOLISM	0.26646768	27	1.652e-06	1.072e-03	ABCB11:103 NR12:201 CYP7A1:281 ABCD3:360 ABCG5:389 VDR:401
CAIRO_HEPATOBLASTOMA_DN	0.09072180	232	2.046e-06	1.206e-03	GCDH:11 GNM1:35 AGXT:45 SDS:55 TMPRSS2:60 SLC1A1:74
REACTOME_ERCC6_CSB_AND_EHMT2_G9A_POSITIV	0.22474241	37	2.251e-06	1.217e-03	H4C2:156 H4C1A:207 H3C13:273 H4C15:325 H2BC15:402 GATAD2A:418
REACTOME_METABOLIC_DISORDERS_OF_BIOLOGIC	0.24410779	30	3.714e-06	1.854e-03	CYP11A1:27 CYP17A1:79 ACY1:104 CYP2R1:264 CYP21A2:565 MAT1A:635
KEGG_VALINE_LEUCINE_AND_ISOLEUCINE_DEGRA	0.20229907	43	4.465e-06	2.069e-03	EHHADH:5 ACADH:132 BCAT2:145 MCCC1:260 HIBCH:287 ACADSB:390
REACTOME_ACTIVATED_PKN1_STIMULATES_TRANS	0.25325329	27	2.588e-06	2.274e-03	H4C2:156 H4C1A:207 H3C13:273 H4C15:325 H2BC15:402 H4C3:448
REACTOME_TRANSPORT_OF_BILE_SALTS_AND_ORG	0.14217444	82	8.697e-06	3.017e-03	SLC14A2:15 SLC6A12:42 SLC6A6:46 SLC13A2:59 RHAG:62 RHCG:200
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.08472924	232	9.200e-06	3.017e-03	SLC5A12:3 SLC14A2:15 SLC4A1:20 SLC6A12:42 SLC6A6:46 SLC13A2:59
CHIANG_LIVER_CANCER_SUBCLASS_PROLIFERATI	0.10564658	151	7.681e-06	3.017e-03	EHHADH:5 GCDH:11 TAT:30 AGXT:45 ABCB11:103 DHTKD1:124
HOSHIDA_LIVER_CANCER_SUBCLASS_S3	0.08447857	235	8.583e-06	3.017e-03	EHHADH:5 SLC23A1:6 CRYAA:10 SLC6A12:42 AGXT:45 SDS:55
KEGG_GLYCINE_SERINE_AND_THREONINE_METABO	0.23007896	31	9.302e-06	3.017e-03	GNM1:35 AGXT:45 SDS:55 SARDH:271 BHMT:289 DMGDH:298
REACTOME_SARS_COV_2_MODULATES_HOST_TRANS	0.24974401	26	1.047e-05	3.233e-03	SNRPD3:66 RPS27A:72 SNRPD1:110 RPS15A:208 RPS17:263 RPS23:276
REACTOME_PHASE_I_FUNCTIONALIZATION_OF_CO	0.15016470	71	1.229e-05	3.623e-03	CYP11A1:27 CYP1A1:47 ALDH1A1:65 CYP2F1:68 CYP2R1:264 CYP7A1:281
BOYALUT_LIVER_CANCER_SUBCLASS_G123_DN	0.18096744	48	1.451e-05	4.092e-03	GCDH:11 SLC6A12:42 GOT2:119 NR12:201 MAT1A:635 DMD:647
REACTOME_HOACS_DEACETYLATE_HISTONES	0.17619456	50	1.644e-05	4.445e-03	H4C2:156 H4C1A:207 H3C13:273 H4C15:325 H2BC15:402 GATAD2A:418
BIOCARTA_NUCLEARRS_PATHWAY	0.23345308	27	2.691e-05	6.982e-03	ABCB11:103 NR12:201 ABCD3:360 ABCG5:389 VDR:401 ABCB1:449
REACTOME_RHO_GTPASES_ACTIVATE_PKNS	0.17934857	44	3.875e-05	9.667e-03	H4C2:156 H4C1A:207 RHOA:238 H3C13:273 H4C15:325 H2BC15:402
WP_ALTERNATIVE_PATHWAY_OF_FETAL_ANDROGEN	0.38992153	9	5.104e-05	1.226e-02	CYP11A1:27 CYP17A1:79 HSD17B3:470 STAR:772 CYB5A:949 HSD17B6:1827
KEGG_SYSTEMIC_LUPUS_ERYTHEMATOSUS	0.13828272	71	5.662e-05	1.267e-02	SNRPD3:66 SNRPD1:110 H4C2:156 H4C1A:207 CIR:246 H3C13:273
REACTOME_ERYTHROCYTES_TAKE_UP_OXYGEN	0.47513113	6	5.562e-05	1.267e-02	SLC4A1:20 RHAG:62 CA4:183 CA2:186 CA1:203 AQP1:1611
REACTOME_ASSEMBLY_OF_THE_ORC_COMPLEX_AT	0.21340731	29	6.979e-05	1.509e-02	H4C2:156 H4C1A:207 H3C13:273 H4C15:325 H2BC15:402 H4C3:448
REACTOME_CONDENSATION_OF_PROPHASE_CHROMO	0.19671466	34	7.226e-05	1.512e-02	H4C2:156 H4C1A:207 H3C13:273 H4C15:325 H2BC15:402 H4C3:448
REACTOME_CYTOCHROME_P450_ARRANGED_BY_SUB	0.18196134	39	8.456e-05	1.662e-02	CYP11A1:27 CYP1A1:47 CYP2F1:68 CYP2R1:264 CYP7A1:281 CYP21A2:565
REACTOME_SIRT1_NEGATIVELY_REGULATES_RRNA	0.20423569	31	8.324e-05	1.662e-02	H4C2:156 H4C1A:207 H3C13:273 H4C15:325 H2BC15:402 H4C3:448
REACTOME_CLASS_C_3_METABOTROPIC GLUTAMAT	0.21372048	28	9.089e-05	1.708e-02	TAS2R10:24 TAS1R2:581 GRM2:731 GRM8:861 TAS1R1:1119 CASR:1421
WP_DRUG_INDUCION_OF_BILE_ACID_PATHWAY	0.31233305	13	9.213e-05	1.708e-02	ABCB11:103 NR12:201 SLC51A:275 CYP7A1:281 SLCOI1:394 VDR:401
CAIRO_LIVER_DEVELOPMENT_DN	0.08007722	200	9.912e-05	1.738e-02	GCDH:11 GNM1:35 MAN2A1:40 TMPRSS2:60 CYP2F1:68 ECH1:123
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.21278893	28	9.753e-05	1.738e-02	CYP11A1:27 CYP1A1:47 CYP17A1:79 COMT:93 SULT1E1:215 CYP7A1:281
WP_DISORDERS_OF_GALACTOSE_METABOLISM	0.31838897	12	1.340e-04	2.288e-02	GYSD:417 PYGL:711 GYS1:784 GYG1:194 GBE1:2012 YG2:2028
WP_ANTIVIRAL_AND_ANTIINFLAMMATORY_EFFECT	0.20792497	28	1.403e-04	2.334e-02	TMPRSS2:60 KBKB:359 NOX1:589 NQO1:663 TNF:998 GCLC:1161
REACTOME_RECYCLING_OF_BILE_ACIDS_AND_SAL	0.27388184	16	1.489e-04	2.415e-02	ALB:9 SLC10A2:36 ABCB11:103 SLC51A:275 SLCOI1:394 SLC51B:808

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Renal salt wasting	0.38265781	15	2.890e-07	2.834e-03	CYP11A1:27 KCNJ1:43 CLCNKA:138 CLCNKB:169 SCNN1A:270 SCNN1B:278
Hyperactive renin-angiotensin system	0.45997966	8	6.632e-06	2.151e-02	KCNJ1:43 CLCNKB:169 SLC26A3:259 SCNN1A:270 SCNN1B:278 SCNN1G:1073
Hyperammonemia	0.11922282	117	8.774e-06	2.151e-02	MANBA:64 NPC1:159 NAGLU:165 CPT1A:176 MCCC1:260 NAGS:269
Increased plasma renin activity	0.45997966	8	6.632e-06	2.151e-02	KCNJ1:43 CLCNKB:169 SLC26A3:259 SCNN1A:270 SCNN1B:278 SCNN1G:1073
Nephrocalcinosis	0.16926377	52	2.455e-05	4.814e-02	CLDN16:1 SLC4A1:20 KCL4:43 AGXT:45 SDS:55 TRPM6:98
Renal tubular acidosis	0.20292370	35	3.289e-05	5.375e-02	CLDN16:1 EHMHADH:5 SLC4A1:20 KCNJ1:43 TRPM6:98 CLCNKB:169
Glycinuria	0.31351807	13	9.094e-05	6.369e-02	SLC6A19:214 ACADM:409 SLC6A20:793 GCSH:1307 PCCB:1501 ALDH4A1:1522
Hypercalcaemia	0.17692976	41	8.942e-05	6.369e-02	CLDN16:1 KCNJ1:43 WNK4:251 SLC12A3:323 VDR:401 PTH:451
Hyperkalemia	0.29080696	17	6.131e-05	6.369e-02	CYP11A1:27 CYP17A1:79 KMK4:251 SCNN1A:270 SCNN1B:278 SCNN1G:1073
Hypochloremia (disorder)	0.14619322	6	8.910e-05	6.369e-02	KCNJ1:43 CLCNKA:138 CLCNKB:169 SLC26A3:259 SLC12A1:1225 BSND:1459
Inborn errors of Metabolism	0.42323217	66	6.489e-05	6.369e-02	GCDH:11 AGXT:45 ACY1:104 HEKX:120 HGD:158 CA2:186
Increased circulating renin level	0.40853532	8	6.299e-05	6.369e-02	CYP11A1:27 KCNJ1:43 CLCNKB:169 SLC12A3:323 SLC12A1:1225 NR3C2:1290
Malabsorption Syndrome	0.19422873	34	8.946e-05	6.369e-02	PNLP1:8 SLC10A2:36 FABP1:90 LCT:108 MTPP:157 CUBN:204
Serum chloride level decreased (finding)	0.46193322	6	8.910e-05	6.369e-02	KCNJ1:43 CLCNKA:138 CLCNKB:169 SLC26A3:259 SLC12A1:1225 BSND:1459
HYPERGLYCINURIA (disorder)	0.28819491	15	1.116e-04	7.292e-02	SLC6A19:214 ACADM:409 SLC6A20:793 GCSH:1307 PCCB:1501 ALDH4A1:1522
Monoparesis - leg	-0.18895854	34	1.383e-04	7.975e-02	HSPB1:29 SPG21:228 NF1:275 SPART:513 FBXO38:692 HSPD1:699
Renal Insufficiency	0.07360574	228	1.371e-04	7.975e-02	CLDN16:1 ALB:9 SLC4A1:20 AGXT:45 CLCNKA:138 CLCNKB:169
Anemia, hereditary spherocytic hemolytic	0.44842946	5	1.731e-04	8.487e-02	ANK1:19 SLC4A1:20 SPTAL:57 EPB42:272 SPTB:708 NA
Benign recurrent intrahepatic cholestasis	0.44536199	6	1.581e-04	8.487e-02	ABCB11:103 ATP8B1:293 GGTLC3:641 GGTCL1:922 GGT1:1095 GGT2:1682
Spherocytosis	0.44842946	5	1.731e-04	8.487e-02	ANK1:19 SLC4A1:20 SPTAL:57 EPB42:272 SPTB:7